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Lesson Notes

MODULE 7 | LESSON 4

FINANCIAL ETHICS

Reading Time	45 minutes
Prior Knowledge	Market Risk, Credit Risk, Market Making
Keywords	Fiduciary Responsibility, Spoofing

1. Introduction to Financial Ethics

For Adam Smith, “a free market must reset on a sound ethical foundation” (The Ethics Centre). Ethics matter. Financial markets exist to the extent that people behave ethically. Recall from Module 1, if a businessman fell behind on his debt, his creditors would break the bench on which they conducted business, giving us the Italian word for “broken bench”: bankrupt. Back then, the industry regulated itself. The scope, complexity, and volume of finance today would bewilder businesspeople from thousands of years ago. Ethical dilemmas exist in areas today that didn’t exist even a generation ago. In this final lesson of the module, we will look at ethics in different categories.

1. Taking Things That Aren’t Yours: This includes plagiarizing and taking code that doesn’t belong to you.
2. Hiding Things That You Shouldn’t Hide: This relates to a lack of internal controls and leads to hiding trading losses that can jeopardize the health or existence of a financial institution.
3. Lying to Colleagues or Customers: This includes providing customers with false prices, fabricating emails that relate to hiding trading losses, or failing to disclose the risks of an investment.
4. Trades That You Shouldn’t Do: These examples include insider trading or spoofing markets.
5. Failure to Disclose Risks: This includes neglecting to act in the client's best interest by failing to disclose risk.

2. Taking Things That Aren’t Yours

It is important to act ethically when “no one is looking.” If you knew doing something was wrong, but believed you wouldn’t get caught, would you still do it? This is an example of an ethical question. One need not be working in the financial industry to face this temptation. This temptation arises each time a

student is faced with what seems like a daunting assignment as the student ponders resorting to plagiarism. Our recommendation—in fact, our school's policy—is not to do it. Of course, most schools say the same; some, including WorldQuant University (WQU), even go to the point of having students sign an honor code asserting that they will maintain the integrity of the institution with statements like "Do your own work" and "Give credit where credit is due."

The "*taking things that are not yours*" issue can present itself even at work. When you work for a company and create software to solve certain problems, unless the company explicitly tells you otherwise in a contract or some kind of agreement, the software you created belongs to the company that employs you. That means that if, for any reason, you no longer worked for that company, you cannot bring that code with you in any form and use it. In particular, the worst possible thing would be to use that code you developed when working for company X to the advantage of company Y. In this case legal actions could be brought against you in certain cases. However, you could not very likely be forbidden to recreate the software line by line from scratch. And this is another reason why simply being able to use code other people developed does not make for valid quantitative analysts since as soon as they lose access to a certain program or are unable to steal it, they are basically unable to perform the tasks asked of them.

Same goes for lists of clients. Those clients belong to the company even if you were the ones contacting them on behalf of the company. Again, evading this obligation could also cause your previous employer to bring legal actions against you.

But let's consider the academic issue of plagiarism. It is possible to outline three different forms of plagiarism that, sadly have become relatively common due to the ease with which people have access to internet.

1. Using appropriate references without citing them.
2. Using inappropriate references—that is, blatantly copying large portions of a previous submission from a previous group or semester.
3. Uploading your own solutions to publicly available course portals.

Regardless of the type, all plagiarism is wrong. Let's take a dive into how plagiarizing equates to stealing, by examining the 10 groups of people that are the victims of this theft.

First and foremost, plagiarists steal from the original source—they try to pass off other people's work as their own. The researcher, author, journalist, or blogger who performed their own hard work, critical thinking, and creativity to produce a journal article, book, or blog post is denied credit. Using that source without citing it is the most classical form of plagiarism. Our recommendation is two-fold: first, be sure to cite it properly within the text and within the references. Second, be sure to add original content that connects the work to the analysis of the particular argument or assignment. If you do this, your work will look more purposeful in that you are deliberately utilizing it to support your own point of view or perspective on the issue at hand.

Second, plagiarists steal from their group mates—specifically, they take away the opportunity for their group mates to earn a passing grade on a group work project. Remember that at WQU, plagiarized assignments earn a zero, and each individual in the group earns a zero, not just the plagiarist. In this

program, students collaborate in the form of a group work project more than two dozen times. One of the most important parts of this educational experience is building teamwork, camaraderie, and trust with one's peers. By plagiarizing, the culprit violates the trust of their peers, and significantly diminishes the opportunity to build connections and engage in fruitful collaboration. Frequently, the group disbands after that happens, and each person's learning experience is weakened. Some plagiarists believe that just the plagiarized portion earns a zero, or perhaps just the person who plagiarized. This is 100% **not** the case. Plagiarism is a poison, tainting the entire submission, not just the portions that have been plagiarized. The easiest way to build animosity and resentment is to plagiarize and damage the grades of one's innocent groupmates.

Third, plagiarists steal from their classmates—specifically, they steal their time and effort by taking a shortcut that gets them to the end goal: a completed project. Suppose a plagiarist spends very little time and effort on a paper, doesn't get caught (a probability to be discussed later), and somehow earns an inflated grade. This is unfair to students who earn their grades through persistence and hard work. Moreover, the plagiarist may be emboldened to brag to students how easy it was, influencing other students who observe the "easy way" and may be tempted to plagiarize as well. Some students who did not do as well may simply question why they spent so much time and effort and still achieved lesser results. This act sets a bad example by claiming results without work.

Fourth, plagiarists steal from the graduates in the program—specifically, they adversely affect the reputation of the program, diminishing the value of a WQU degree in the marketplace. Again, suppose a plagiarist somehow managed to get through the Risk Management class and earn a good grade. Imagine that person goes on an interview and gets questioned about certain topics but is unable to provide informed or competent answers. This is plausible because the student never had to do the hard work to learn the concepts during their studies. The plagiarist may end up embarrassed by the interview, but employers may quickly begin to associate the program with inflated grades and underqualified students. Consequently, employers will become discouraged from interviewing or hiring other graduates from the program. This theft is serious because it creates a real opportunity cost for students who worked hard and followed the rules.

Fifth, plagiarists hurt future students. How? When they upload course notes to a website, plagiarists create a temptation for those future students to plagiarize for themselves. University policy forbids this, but the real damage again is to students who follow the rules. Stored files weaken the reputation of the overall program. Uploading solutions tempts students in the future to use previously completed work rather than build their own skills. Just because a file is uploaded doesn't even mean that it is correct! Never upload course content or completed assignments to a portal.

Sixth, plagiarists steal from their professors—specifically, their time and efforts. Teaching faculty truly want students to learn the material the right way—otherwise, they wouldn't be teaching. Faculty are happiest when students learn and grow. Grades are not meant to be a prize but a reflection of a student's dedication and performance. When a student plagiarizes, they attempt to replace that dedication with someone else's work passed off as their own. It undermines the efforts of content developers, reviewers, and editors; lecturers; graders; and other faculty members who provide materials, share experience, and offer feedback for the growth and development of students.

Seventh, plagiarists steal from the administrators of the school. The mission of the university is “to make advanced, quality education accessible to capable students everywhere by leveraging technology at scale to deliver entirely free online programs.” It does so at no cost to the student. The existence of this school is a testament to how the founders and administrators want to make an accredited master’s degree in financial engineering widely available to all by removing hindrances due to cost. While the school does not require money, it does require integrity. It makes no sense to steal from the institution that has invested resources and personnel in you and your professional development by diminishing the school’s brand in the marketplace. Plagiarizing is an affront to the university by threatening its ability to fulfill its mission. This is the reason why plagiarists are dealt with harshly. Please respect the mission of the university.

Eighth, plagiarists steal from their future employers—specifically, they are using deception to take a position for which they may not be qualified. A future employer assumes that the student has completed the rigorous financial engineering workload required for a two-year master’s program. This comprehensive qualification is one that most people in the industry don’t achieve. Even if the plagiarizing employee continues to do good work for their employer, their work is predicated on a foundation of inflated, or even fabricated, skills and competencies. People often get fired if they were found out to have plagiarized.

Ninth, plagiarists steal from their future coworkers. By presenting an embellished skill set on a team, plagiarists pretend to be something they are not. Coworkers will expect the employee in question to have the required skills yet may notice that person engaging in deflection, postponement, delegation, or any other method to avoid doing work they are supposed to know. The longer the plagiarist is an employee, the more difficult it is to come clean and admit they don’t know something. This may even embolden the plagiarist to continue lying or stealing. This creates the problem of false pretenses. Here’s a well-publicized example. The Dean of Admissions at MIT, one of the most prestigious technical schools in the world, lied on her resume (Martin). After a 28-year career at the school, the lie was uncovered and the Dean was fired, wiping away decades of her good service, which was all built on false pretenses. Reputations are like credit risk: it is difficult, if not impossible, to bounce back from a “default” state.

Finally, plagiarists steal from themselves. Consider a student starting out in the program. They will dedicate some 1,500 to 2,000 hours to get a master’s degree. Once they earn a master’s degree, they have it for life! (Some certifications must be renewed periodically.) This will become part of their professional profile. If they think about how relatively small any one assignment or project is in the scope of the entire master’s program, they will not be tempted to plagiarize. The student realizes that if they do plagiarize, potentially all their effort could be for naught. Why? Because a single violation of the Academic Integrity policy due to plagiarism earns its culprit a failing grade for the class or termination from the program. Is one assignment worth it?

Plagiarism in this day and age extends beyond copying from some external source. The introduction of ChatGPT and similar AI tools has made it possible to avoid doing the work by simply asking the tool the same questions that were asked of students and then copying and pasting the answers. Although the use of AI tools is not forbidden while a student is learning certain materials in a class, it is equivalent to

cheating if those tools are utilized during tests (such as graded quizzes, group work projects, etc.). WQU takes this very seriously, and heavy penalties are imposed when AI tools are used.

We'll end our discussion of plagiarism with a comparison to credit risk modeled via binomial trees. Suppose a business has to pay back a debt. There are two outcomes: they can either pay it or default, as illustrated by this binomial tree.

Business owes debt. Either the business pays or defaults.

Default is an absorbing state. This means that once the business defaults, it stays in a default state. They would declare bankruptcy and be forced to restructure or go out of business.

Now consider a student who has an assignment due and is considering plagiarism. There are two outcomes: they can either do their own work, or they can plagiarize, as illustrated by this binomial tree.

The student owes an assignment. There are two paths:

- *Does the student do their own work? This leads to a grade based on quality of work.*
- Does the student plagiarize? This leads to different grades.
 - Grade of stolen material (not necessarily an A)
 - 0 if caught—*not only for the cheater, but for their group as well.*

If students do their own work, they learn and grow and receive a grade for their results. If the students plagiarize, they bypass the learning and growing phase and receive a failing grade for the course.

What do these two problems have in common? The probabilities are *unknown*. One of the difficulties in credit risk is that default is relatively rare. In this program, we will spend time studying how to calculate the default probabilities by looking at stocks, corporate bonds, credit default swaps, and credit ratings. Similarly, one of the difficulties in *deciding* to plagiarize is estimating the probability that you will get caught. Students may not estimate at all, thinking "they will never get caught." Ironically, this lesson is one that will be taught in Behavioral Finance, underestimating probabilities when it applies to them personally.

One thing to know is that the university is equipped with cutting edge tools for detecting plagiarism. For example, Turnitin is incredibly effective at finding plagiarized content. Indeed, students must run their assignments through Turnitin before submitting them. The list of sources that Turnitin checks is constantly expanding. Therefore, the potential plagiarist may think they know the probability of getting caught, but it is likely that they underestimate this.

Moreover, the student who plagiarizes tends to *underestimate the loss* given they are caught. When they inevitably get caught, every plagiarist is sorry after the fact. The excuses tend to include some combination of "I'm sorry," "I didn't know," and "I won't do it again." While the apology may seem or even be sincere, the plagiarist wanted a good grade. This phenomenon is another variable in credit risk known as the LGD: Loss Given Default or the recovery rate (1 minus the LGD). When the revealed plagiarist negotiates for just a few points off rather than an official 0, they are attempting to set their recovery rates. For example, they plead, "Can you just take the 10 points off on the question I plagiarized instead of the whole assignment?" As we previously addressed, plagiarism is poison that

taints the whole assignment. The plagiarist should expect a 0 on their assignment, which amounts to a recovery rate of 0. Furthermore, many faculty will not write letters of recommendation or work with students on their Capstone project if the student has a history of plagiarism. One's reputation is one's brand. Our advice is to maintain the brand's quality by working professionally.

There's even another variable in credit risk that relates to plagiarism: default correlation. Someone who plagiarized one time is likely to have done it before—and to do it again. In other words, the plagiarist's behavior is likely to be correlated across assignments and classes.

Maintaining good financial ethics is one of the program outcomes. As a student, avoiding plagiarism is one of the best ways to build these ethical practices. Now, let's look at other examples in industry where ethics can be an issue.

One other example of "taking things that aren't yours" comes from industry. Sergey Aleynikov stole high-frequency trading software from his employer Goldman Sachs. Then, he quit and accepted an offer at a hedge fund. Stealing the code triggered security alerts, which notified his employer of his theft. He was caught, tried, convicted, and served time in jail (Henning). There is much more to this story, which became the basis of the best-selling book *Flash Boys* by Michael Lewis.

Yihao Ben Pu did the same thing: He stole code from his employer and attempted to bring it to another hedge fund. He was also caught, tried, convicted, and served time in jail (USAO).

Stealing code from one's employer may not seem like a crime, especially if that person wrote the code themselves. But a careful reading of an employment contract will show that the code belongs to the company rather than the employee. In fact, this code can be one of the most valuable assets a hedge fund has if it describes trading strategies. Our advice is to not take what is not yours, whether it is someone else's assignments or code.

3. Hiding Things That You Shouldn't Hide

A second set of ethics addresses "hiding things that you shouldn't hide." Our first example is a classic one because it occurred nearly 30 years ago and is often the first case study in risk management for financial crises. It was also turned into a Hollywood movie (see the 1999 movie *Rogue Trader* starring Ewan MacGregor). Barings Bank employee Nick Leeson made inappropriate derivative trades and hid the losses, to the point that the catastrophic size was enough to bring down the 230-year-old financial institution (Smith). Sometimes, the losses are hidden not by a single individual, but by multiple people in an organization. For example, Daiwa Bank admitted to hiding more than \$1 billion in trading losses (Truell). This loss was significant but did not cause Daiwa to go bankrupt. People at the firm pled guilty, causing the firm to pay a record fine of \$340 million.

Risk management was in place at all these institutions, but a combination of unethical behavior and negligence contributed to these becoming epic financial disasters. This is not merely market risk—it is the lack of controls, checks, and balances to protect the firm from its own unscrupulous employees.

4. Lying to Colleagues or Customers

A third set of ethics addressed lying to colleagues or customers is wrong. Lying to colleagues may not seem so bad, but what if those lies were to hide losses? Jerome Kerviel at Société General lost \$7.2 billion in trading losses. This was 1.5 times the market capitalization of the bank itself! He had hacked the company's surveillance system to bypass the proper risk channels. He traded enormous sizes without taking the corresponding hedges. As he lost money, he fabricated emails that made it look like his positions were being monitored and managed appropriately. Eventually, he was caught, tried, convicted, and sentenced to prison. However, he was also fined for the amount he lost: some \$6.7 billion dollars. Clearly, he doesn't have that kind of money, but he remains in debt to the bank for that amount (Ayandayo).

When it comes to customers, there is much more formal regulation in place. Oftentimes, traders, asset managers, brokers, or dealers must act as a **fiduciary**, which means the agent acts on behalf of a client to serve the client's best interests. Sometimes, advisors lie to their clients because they have huge losses on their customer's accounts, but don't report these directly to the client. This could take the form of a Ponzi scheme (as in the Madoff example), simply losing money and not telling anyone (for one example, see Longo), or simply using client funds to subsidize a lavish lifestyle until one is caught. If you google the words "advisor," "using client funds," and "lavish lifestyle" (without quotes), you will find thousands of articles of offenders. This relationship is built on trust. Breaking that trust violates the fiduciary responsibility.

5. Failure to Disclose Risks

Another group of ethics violations occurs when financial professionals fail to disclose risks. Ethics can break down when a trader, dealer, or advisor fails to work in the best interest of the customer. These are well explained in the interesting article by Claire Hill: "Bankers Behaving Badly."

This article addressed different types of cases. The first type of case is where advisors sell unsuitable investments to clients. A well-known example involves former Goldman Sachs banker Fabrice Tourre, who sold complex securities to customers who were not suitable for these products, failed to disclose the proper risks, and did not work on behalf of clients (U.S. Securities and Exchange Commission, "Goldman Sachs to Pay"). Another type of case is where advisors and clients both work unethically to deceive or defraud a third party. The interest rate swaps disaster in Jefferson County is a classic example (U.S. Securities and Exchange Commission, "J.P. Morgan Settles"). Representatives of a Wall Street firm were charged with making unlawful payment to county commissioners to agree to a deal that benefited the bank and the commissioners personally. However, the transaction led to the bankruptcy of a county—one of the largest municipal bankruptcies in the entire United States. The bankers did not disclose the risks and paid off the commissioners, who accepted the higher than usual rates on the swap payments. Think of this as a two-stage fiduciary responsibility, in that both the bankers and the commissioners did not act in the best interest of their constituents, respectively. Fines were paid and jail sentences were served. These types of swap disasters are unfortunately all too common, and a book was written about them (see Abdel-Khalik's *Brazen: Big Banks, Swap Mania, and the Fallout*).

Other times, failure to disclose risks can help cause a firm to go under. See the Jeffers article on how failing to disclose the risks of repo transactions caused Lehman Brothers, a 100+ year old company, to

go bankrupt during the Global Financial Crisis. Thanks to regulation that arose out of the Enron Crisis (Sarbanes-Oxley), measures were in place to show how the accounting was fraudulent.

These types of unethical salesmen helped motivate the expression "caveat emptor," let the buyer beware. Prudent financial professionals should know the risks of the transactions they enter on behalf of their constituents, clients, and companies.

6. Trades You Shouldn't Do

In this last section, let's look at "trades that you shouldn't do." The first is insider trading. Insider trading means trading on information that is not materially public. Mathew Martoma was sentenced to nine years in prison for insider trading that involved approximately \$275 million in illegal profits and avoided losses. Interestingly, this was not the only dishonesty in his history. He had been enrolled in Harvard Law School but was expelled for grade manipulation and dishonesty. Later, he attended the Stanford Graduate School of Business and earned an MBA. However, when Stanford learned that he had failed to disclose his Harvard expulsion, Stanford revoked his MBA. "False pretenses" has no expiration date, as we saw earlier for the MIT Dean of Admissions. This means that the threat of being exposed for plagiarism or other lies can ruin your reputation and work that was done earnestly and honestly.

An even more famous case than Mathew Martoma is Martha Stewart, who was arrested, convicted, and served time in prison for insider trading (U.S. Securities and Exchange Commission, "SEC Charges Martha Stewart").

Aside from insider trading, another unethical trade type is spoofing—something Navinder Sarao did on May 6, 2010. He placed unthinkable sizes of a short in the S&P 500 E-mini futures. This is known as spoofing the market—placing orders with no intention of getting executed, but merely influencing the market to affect prices in a way that benefits you. This trade triggered an enormous market selloff that became known as the Flash Crash. This crash temporarily wiped out approximately \$1 trillion (million million) dollars in market value. Sarao was found guilty of spoofing the market (Reuters). His behavior exposed a vulnerability in the financial exchanges. This case is a classic example of systemic risk.

7. Conclusion

What are the costs of unethical behavior in the financial marketplace? Likely, these costs are immeasurable. We can say that these behaviors add uncertainty to challenges like credit risk, financing costs, volatility, correlation, leverage, non-linearity, and liquidity and can exacerbate model failure and crises. It makes everything more difficult due to the broken trust implicit in transactions. Please read the excellent article by Svetlova about details on practices of good ethical behavior.

Finance is a highly leveraged field where the temptation to make significant amounts of money in an unethical manner is prevalent. If ever in doubt, ask yourself: would a jury of my peers consider my behavior appropriate? Warren Buffet, one of the greatest investors of all time, gave the following advice, called the newspaper test (Schwantes). Suppose a competent but unfriendly reporter wrote an article about a decision or action a businessperson made. Everyone reads it. How would the businessperson feel about it? If the decision passes that test, it's okay. In other words, if a decision you make becomes

public and you feel even remotely uncomfortable or embarrassed that people know about it, it's probably not a good decision. Buffet adds, "we can afford to lose money, but we can't afford to lose reputation" (Schwantes).

The take-home message is very clear: Ethics matter! Financial markets depend on it. Everything you will learn, the value you add, and the ladder you climb are all thwarted by poor ethical behavior. The damage to one's reputation is lasting. Reputations are asymmetric: decades of hard work in building a great reputation can quickly and easily be undermined by a single unethical act that takes moments to perform. Nurture your reputation by learning from the mistakes of others and committing yourself to acting with integrity in your career.

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